Data Science

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This is meant to be a thorough guide to the Data Science job recruitment process, from before you send your first application to when you accept an offer. Feel free to skip around to parts you feel you need the most help in, or read from beginning to end.

Good luck!

0. Getting Started

▼ 0.1 — Job application timelines

Big Tech

- Late August - Early December
- Most roles are usually finalized by the end of the year, but can go into February sometimes

Mid-Small Tech

- Late August - Mid-March
- These are usually more flexible, but hiring pretty much hard stops around April (for summer internships)

Startups

- Apply anytime, either by sending in an application or by DM/emailing someone that works there; also, there are no hard deadlines on startups

▼ 0.2 — Getting ready to apply

▼ 0.2.1 — Fix your resume

- Resume formats are pretty up to preference, but there are a few important parts:
  1. Make it *human readable*
     Fonts shouldn't be hard to read or too small (should be size 10-12)
  2. Make it *computer readable*
     If you used something like Microsoft Word, Google Docs, OverLeaf, etc. to create your resume, don't worry about this. All this means is if you open your resume as a PDF, you should be able to select text.
  3. Keep it *consistent*
     Your resume formatting should be consistent throughout the whole thing. Fonts for jobs should all be the same, your bolding/italicizing shouldn't be in random places, etc.
- If you aren't sure where to start, ask any DSS member you want and they should be able to help you start.

▼ 0.2.2 — Create a cover letter template

- This is kind of a controversial subject when applying to jobs- some people prefer to send in a cover letter whenever possible, and others don't.
  - The general consensus online is that cover letters don't matter/don't do a whole lot in tech, but some people swear by sending them in.
- If you *are* going to write cover letters, create a template and use it whenever you are asked for one. Here are some links:
  1. https://medium.com/@madelinemann/ive-read-1000-cover-letters-and-this-is-what-will-get-you-the-interview-b534660d6fd5
  2. https://www.reddit.com/r/datascience/comments/tag8l5/my_guide_to_writing_a_killer_cover_letter/
  3. https://resumegenius.com/cover-letter-examples/technical-cover-letter

▼ 0.2.3 — Sign up for a Simplify.jobs account

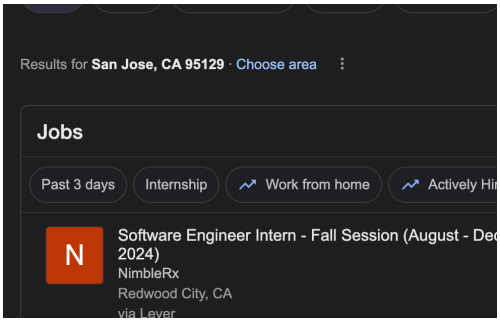
- Simplify will automatically fill out Workday, icims, and most other job application platforms automatically for you
- It's boring to do but you'll only have to do it once- copy & paste your resume (yes all of it) in the "My Profile" section. This way, it will automatically fill for you in the future.

▼ 0.2.4 — Set up LinkedIn job alerts

- If you're scrolling LinkedIn jobs, usually you'll find something with 1000+ applicants that was posted 2+ weeks ago. This is *not* giving yourself the best shot.
- Instead, set up alerts to get notified about jobs *when they come out*.
- Follow this guide: How to set up LinkedIn Alerts for specific jobs

▼ 0.2.5 — (Optional) Set up Google job alerts

- For the same reason as the LinkedIn job alerts
- How to set it up:
  1. Search <job_name>
     e.g. "Software Engineer Intern"
  2. Click on "Jobs" in the results



  3. At the bottom left of your screen, turn on "new job alerts"

▼ 0.3 — What to expect

Here's an outline of the whole process. For details, read further in this guide. This may not represent *every* interview process, but this is usually how it goes for Data Science roles.

1. **Sending in your application**
    - This is the worst part of the process (really). Do all the steps in 0.2 before applying
2. Online Assessment
    - Usually a Hackerrank, Hirevue, or a take-home project.
3. Interview #1: Behavioral Interview
    - Most commonly with a recruiter, could be with a technical person at a smaller company
4. Interview #2: Technical Interview
    - Topics can vary, usually with a data scientist or data science manager
5. Offer call
    - Congrats! You got through the interviews and got an offer.
6. Accepting the offer
    - Where you formally take the job, ending your job search journey.

1. Applying to Jobs

▼ 1.1 — Where to find jobs

- It's pretty up to preference where you want to find jobs, but for DS roles usually a combination of LinkedIn and Handshake does the trick. You can never apply to too many roles, so feel free to use others if you have the time.
- Commonly used sites:
    - LinkedIn - Data Science Internships - LinkedIn Job Search, filtered to "Data Science Intern"
    - Handshake - Data Science Internships - Handshake Job Search, filtered to "Data Science Intern"
    - Data Science List - Internships/Jobs - not a comprehensive list, but has some good ones
    - Indeed - Data Science Internships - Indeed Job Search, filtered to "Data Science Intern"

▼ 1.2 — How many to apply to

- You can never apply to too many jobs, but this really depends on how much time you are willing to put in.
- At minimum, you should try to apply to 10+ jobs per day during recruiting season, but ideally do something like as many as you can in an hour or two every single day (which ends up being much more than 10).
    - However, it's better to be consistent (like applying to 10 per day) instead of going crazy on one day and burning out (like applying for 100 on a random Tuesday and not applying for a month)
- Something interesting that you may run into as time goes on:

- It's possible that you may start "running out of jobs" to apply to. By this, I mean you may start seeing the same job listings on LinkedIn you've already applied to even though you refresh or change filters.
- This is annoying, but a good problem to have as it means you've applied to a lot of jobs. There's no fix for this as far as I know, just keep calm and keep applying.

2. Interview Prep

▼ 2.1 — General Tips

This is the main part of the job search process, so the main thing to remember is that you won't be perfect when you start get going. You might fail your first few online assessments, fumble your words on a "Tell me about yourself" interview question, or get super nervous before an interview.

The hardest part is staying consistent in your prep for interviews, as there is quite a bit to do. Take it one-by-one so you aren't overwhelmed by the many things to do, and figure out a routine that works best for you.

Stay motivated! Recruiting is a numbers game, especially in today's job market. You go to UC Berkeley and are in the BEST organization for Data Science. You are absolutely qualified, now it's just a matter of showing off your skills (maybe reviewing them too lol).

Also, remember that others in DSS are here for you when you're preparing for these. If you have questions, ask anyone you would like. If you're looking for a buddy to block out time to study with or practice interviews, just reach out!

▼ 2.2 — Online Assessment (OA)

- Before you read, a couple things to note:
  - HireVue is usually only for Finance or Consulting companies. Most tech companies will **not** use HireVue. If you've never heard of HireVue, click below to learn more.
  - With Hackerrank assessments, this greatly depends on the company. Some companies like JP Morgan Chase will send a regular Leetcode-style Hackerrank question to solve, while others may send a SQL Hackerrank question. Your mileage may vary, so be prepared for either.

▼ 2.2.1 — HireVue

- HireVue is pretty much a platform for one-way interviews used most commonly by finance and consulting companies. Most tech companies **will not** have a HireVue, but you should be ready to do one if necessary.
- These are annoying in the beginning, but with practice you'll see that most questions are basic and usually pretty similar.
- It usually goes something like this:
  1. Watch a 30 second clip of some corporate person read a message thanking you for applying to the job, telling you to come off as geniune as possible, and to answer the question to the best of your ability
  2. You'll be shown a question on your screen and have usually between 1 minute and 3 minutes to prepare (you aren't being recorded here).

3. If you want, you can press “record” early and starting speaking. Otherwise, when time expires, HireVue will automatically start recording you. This is where you answer your question outloud and speak to the camera.
    - It’s *obvious* when you read off of a Google Doc or a Notes app, so do your best to have these memorized if possible, or just look at your notes for keywords to remember
    - Responses are usually capped at 1 min - 3 mins, so HireVue will automatically stop recording when that time is up. If you finish early, you can press “Stop Recording” yourself.
4. You’ll usually have to answer between 2 and 5 questions, which could vary in length for how long you’ll have to prepare or speak.
5. After the last question is answered, HireVue will automatically submit your responses. Then you’re done!

- Kinds of questions you could be asked are usually classic interview questions:
    - Why <company>?
    - What is your greatest strength and your greatest weakness?
    - Tell me about a time where you struggled
    - Tell me about a time when you led a team to overcome a challenge
    - Tell me about your most rewarding job experience
    - And more! Just Google for this or search on Glassdoor

▼ 2.2.2 — Hackerrank

- Some companies may send a Hackerrank to complete after you send in your resume
- Usually you are given a period of 48-72 hours to complete the assessment, with some companies allowing as long as two weeks. Complete it ASAP, don’t wait a week. Ideally finish it within 48 hours.
- Most Hackerrank assessments are some mix and match of Leetcode-style Data Structures and Algorithms, SQL, Python, and Machine Learning.
    - For resources to prepare, go to the “Technical Interview” section below as these will be the same links.

▼ 2.3 — Behavioral Interview

▼ 2.3.1 — Recruiter Call

- This is usually the first live interview you’ll do after the resume round (if no HireVue)
- Usually pretty chill, just make sure you come off as cheerful and bright-eyed as possible
- Know your resume inside and out, be able to talk about *anything* you put on there.
- Be ready to answer classic interview questions just like the HireVue
    - Why <company>?
    - What is your greatest strength and your greatest weakness?
    - Tell me about a time where you struggled
    - Tell me about a time when you led a team to overcome a challenge
    - Tell me about your most rewarding job experience
    - And more! Just Google for this or search on Glassdoor

▼ 2.4 — Technical Interview

- Something that makes the Data Science job application process annoying sometimes is how *variable* the process may be for different companies
  - Some companies may only ask SQL and conceptual ML, others only Python, some maybe only Leetcode-style Data Structures and Algorithms questions.
    - Your best bet is to be prepared for as many of these as possible (obviously, but it had to be said).
- Related Berkeley classes will also be mentioned with these skills, so either review the skills if you've taken the class, preview the content if you are planning on taking it, or add it to your plan if you think you want to take it in the future.

▼ 2.2.1 — SQL

- Related classes: DATA 101, CS 186, DATA 100
- Prepping for SQL isn't too complicated and usually you won't be asked anything overly difficult. From most DSS peoples' experiences and the internet, it seems that most interviews for internships will ask easy-medium level questions (from LeetCode or Hackerrank).

  ▼ This is what one Reddit user said, which may be a bit more than you would need in most interviews or assessments.

  > "In addition to the obvious basics like group by, joins, etc., I'd refresh on window functions, as they're insanely useful and tend to separate intermediate SQL writers from beginners. Know when to use them and how they can make your queries shorter by avoiding self joins to subqueries.
  >
  > Also the basics of data modeling, at least star schemas.
  >
  > And a little about query optimization. A question I usually ask in interviews is: What do you do if your query (though logically correct) runs too long or runs out of memory?
  >
  > Finally, creating tables either via DDL or with select into / create table as. And the various types of inserts and updates. E.g. a common pattern with time series data is to restate old records once per day with some rolling window (say trailing 7 day) so that slight changes in historical data are captured. To do this, you often need to structure the table deliberately with the right partitions based on whatever grain you want to overwrite at.
  >
  > Just know one engine really well - it's unlikely they'll count it against you if e.g. you can't remember the name of a specific date function in MySQL but you know it in PostgreSQL."

- Sites to use:
  - PGExercises - If you're just getting started, this will walk you through the basics of PostgreSQL (a version of SQL) and get you practice with the foundations
  - Hackerrank SQL - This is a more comprehensive list, also better for beginners. You can filter by difficulty, subject, and whether you've solved it or not.
  - DataLemur - SQL & Data Science Practice (similar format to LeetCode) created by Nick Singh, has lots of SQL questions with some ML/Stats questions mixed in
  - LeetCode SQL - A relatively new feature on LeetCode, they've added some SQL questions as well. Doing the 50 questions may be useful.

▼ 2.2.2 - Pandas

- Related classes: DATA 100

- DATA 100 level-skill in Pandas should suffice for most interviews. I would recommend reviewing DATA 100 content and solving a few Pandas question from Leetcode (linked below) to best prepare.
- LeetCode (PANDAS) - new feature, helpful for Pandas/data analysis questions

▼ 2.2.3 — Probability/Statistics

- Related classes: DATA 8, DATA C88S, DATA 100, STAT 134/DATA 140
- The types of questions related to probability and statistics can vary widely, so it's pretty hard to determine what exactly to prepare for.
  - To start, you should know the basic probability/statistics concepts from DATA 8 and DATA 100
  - If you have time, you can review DATA 140 content (if you've taken it) or preview it if you have not. Most of the time, you won't get asked something harder than what would be found in DATA 8/100.
    - Something I (Michael) have been asked a few times is about *expected value* and *conditional probabilities*. Very basic, so make sure you don't fumble those.

▼ 2.2.4 — Conceptual Machine Learning

- Related courses: DATA 100, DATA 144, CS 189, INDENG 142A
- Knowing the models covered in DATA 100 (at least the more recent versions of the course) have been pretty sufficient in most interview situations for me (Michael)
  - Review conceptually how all the models (from linear regression to random forest) work, and if you have time learn the code implementation for them in scikit-learn. If you are experienced and know how to implement them in other packages like Tensorflow or PyTorch, by all means, practice that.
- DataLemur has some ML questions, which can be useful to take a look at for practice.

▼ 2.2.5 — Data Structures & Algorithms (DSA)

- Related classes: CS 61B, CS 170 (overkill for DS)
- Hopefully, you won't see much of DSA in your job search. DSA is less commonly seen for DS roles which is why it is last of the technical skills listed.
- See 💻 SWE for tips and tricks at getting good at DSA (referred to as Leetcode).
- LeetCode 75 - do Easy/Medium DSA problems (not always required, but necessary for OAs at big companies)

3. Navigating Offers

▼ 3.1 — Accepting your offer

Congratulations! If you've gotten an offer and you intend to take the job, this step is quite simple. You'll usually have to call or email the recruiter and let them know you would like to accept the offer.

You'll also probably be sent a DocuSign with forms to sign to get it set in stone. At this point, you've gotten the role. You did it!

▼ 3.2 — If you are waiting on other final results

The safest thing to do when you get an offer is to take it (obviously), but you could be in a situation where you did the final round for two companies within a day or two of each other and want to hear back from both.

If you get an offer from one company but would rather take the other company you haven't heard back from yet, **don't** reject the offer from the first company outright. There is a chance you could rejected from that

second company, so play it safe.

The best thing to do in this scenario is act very enthusiastic and thank your recruiter for the opportunity, and ask them if there is a deadline for you to make a decision by. Usually they will give a time range between 48 hours and 7 days, but keep in mind until you accept an offer, it is **NOT** yours yet. Especially in this economy, accept first, renege (change your mind) if you must.

Old Resources

Recruitment Process (By Pranati)